

Exercise Sheet 2.2

Lecture “Cybersecurity”

Summer Semester 2026

Practical Exercise: Data Encryption Standard (DES)

Note. The S-boxes required for the calculations can be found starting on page 17 of the NIST standard.

Exercise 1: Computation of the First DES Round

Aufgabe 1.

In this exercise, the first round of DES is computed step by step. Let the plaintext bits x_i (indexed from $i = 1$ to $i = 64$ from left) be:

$$x_i = 0 \quad \forall i \in \{1, \dots, 64\} \setminus \{15\}, \quad x_{15} = 1.$$

The key k and all round keys k_i consist entirely of zeros.

- a) State the initial state x before applying the initial permutation IP (4 rows of 16 bits each). x :

- b) What is the output of the initial permutation IP , split into the two halves L_0 and R_0 ?

IP

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Abbildung 1: Initial Permutation

L_0 :

R_0 :

- c) Apply the expansion function $E(R_0)$ and state the inputs of the eight S-boxes. Take into account the XOR operation with the round key k_1 (all zeros).

E BIT-SELECTION TABLE

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

Abbildung 2: Expansion Table

S_1 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>	S_5 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>
S_2 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>	S_6 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>
S_3 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>	S_7 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>
S_4 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>	S_8 : <table border="1" style="display: inline-table; width: 100px; height: 20px; border-collapse: collapse;"></table>

- d) Determine the output of the S-boxes before applying the permutation P . S_{out} :

e) State the output of the permutation P .

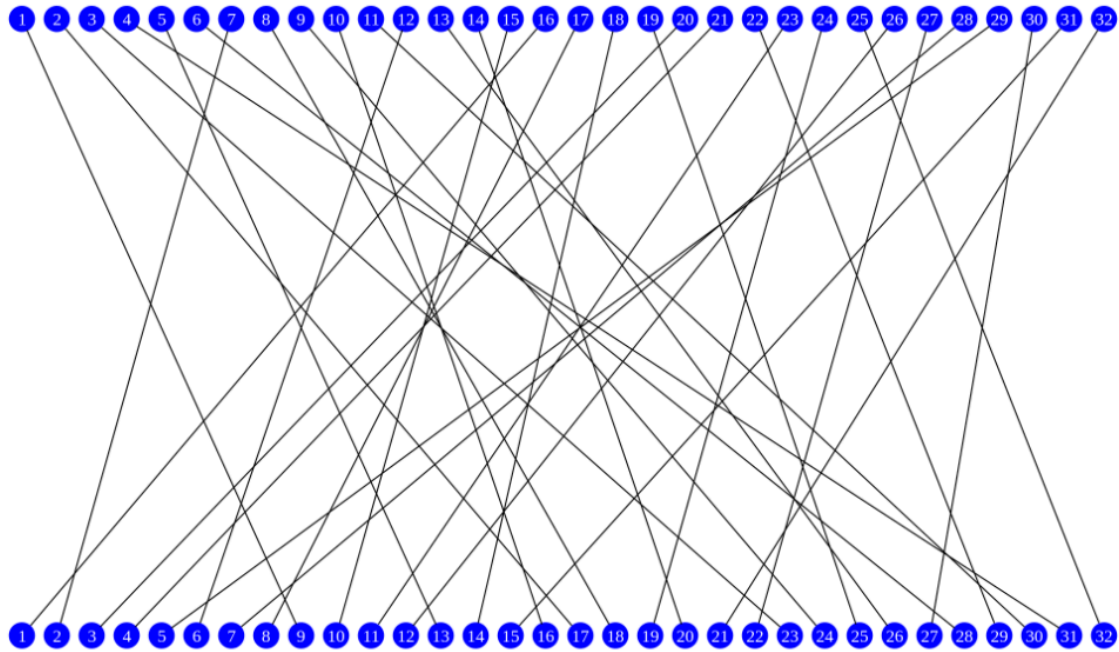


Abbildung 3: Permutation

 $P_{\text{out}}:$ [illegible]

f) Determine the state (L_1, R_1) at the end of the first round. L_1 :

[illegible] $R_1:$ [illegible]

Exercise 2: Avalanche Effect in DES

Aufgabe 2.

For a good block cipher it is desirable that changing a single input bit causes as many output bits as possible to change. This property is also called diffusion or the *avalanche effect*. In the following we examine the diffusion property of DES. We consider a bit sequence as input to DES in which the bit at position $x_{37} = 1$ (indexed from x_1 to x_{64}) and all other bits are equal to 0. The 56 key bits (and therefore all round keys) are all set to 1.

- a) Which S-boxes are affected by bit x_{37} in the first round? What do the input bits of the eight S-boxes look like?

S_1 :		S_5 :	
S_2 :		S_6 :	
S_3 :		S_7 :	
S_4 :		S_8 :	

- b) State the result (L_1, R_1) after the first round. L_1 :

R_1 :

- c) How many bits differ compared to the computation with input $x = 0^{64}$ (same key configuration)? What conclusion can be drawn from this regarding the avalanche effect?

Exercise 3: DES Decryption

Aufgabe 3.

In DES decryption, the same algorithm as in encryption is used; only the round keys are applied in reverse order. This property follows directly from the symmetry of the Feistel structure. We continue the computation from **Exercise 2**: after one encryption round with $x_{37} = 1$, all remaining plaintext bits equal to 0, and a key consisting of ones, the state (L_1, R_1) is given. We now show that an *inverse Feistel round* with round key $k_1 = 1^{48}$ exactly reconstructs the previous state (L_0, R_0) .

- a) Explain why DES decryption can use the same algorithm as encryption. In what order are the round keys applied?
- b) Derive the recurrence of an inverse Feistel round that computes the state (L_{i-1}, R_{i-1}) from (L_i, R_i) and k_i . Apply it to (L_1, R_1) from Exercise 2 with round key $k_1 = 1^{48}$ and enter L_0 and R_0 . L_0 :

R_0 :

- c) Compare the computed pair (L_0, R_0) with the state obtained after applying the initial permutation IP to the original plaintext from Exercise 2. What conclusion follows regarding the correctness of DES decryption?

Summary

- DES was put out to tender in 1972, standardized in 1977, and is based on a 16-round Feistel structure with an effective key length of 56 bits.
- The round function combines expansion, key addition, nonlinear S-boxes, and a bitwise permutation.
- The S-boxes are the only nonlinear components and therefore central to the security of the scheme.
- Diffusion is achieved through permutations and the repeated application of the round function; after just one round, a single input bit already affects multiple output bits.
- Due to the short key length, DES is considered insecure today; in practice, 3DES (transitionally) or AES are used instead.

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